Project Interim Submission: Question Answering on Tabular Data

The LTRCs (Team #5)

Abraham Paul Elenjical 2022114007
Vemula Saketh Reddy 2022114014
Kavuri Vivek Hruday 2022114012

Problem Statement

The main task proposed in SemEval Task-8 is Question Answering (QA) over tabular data using the DataBench [1] benchmark. The objective is to develop models that answer questions about real-world table datasets, where answers can be numerical, categorical, boolean, or lists. The task consists of two sub tasks:

- **DataBench QA**: Answer questions using the full dataset provided.
- **DataBench Lite QA**: Answer questions using a sampled subset of the larger dataset (maximum 20 rows).

The models developed must solely rely on the provided dataset to generate answers.

Exploratory Dataset Analysis

The Dataset we are using is the DataBench benchmark. After exploring the dataset, we found that it includes 65 publicly available datasets across 5 domains - Business, Health, Social Networks and Surveys, Sports and Entertainment and Travel and Locations.

Overview of the Domains

- **Business:** This domain has 26 datasets with approximately 11.6M rows and 534 columns. This includes financial data, real estate information and customer behaviour.
- **Health:** This domain has 7 datasets with approximately 98K rows and 123 columns. This domain includes public health data like stroke prediction and heart failure.

- **Social Networks and Surveys:** This contains a set of 16 datasets with approximately 11.9M rows and 508 columns. This has a focus on social media data and survey data.
- **Sports and Entertainment:** This has 6 datasets in total with approximately 399K rows and 177 columns. This has data ranging from FIFA player data to IMDb movie datasets.
- **Travel and Locations:** This has 10 datasets with approximately 427K rows and 273 columns. The data in this domain contains data such as Airbnb listings and hotel reviews.

Types of Columns

- **Numbers:** 788 Columns - Fares, ages and prices etc...
- **Categories:** 548 columns - Names, classes and categories etc...
- **Dates:** 50 columns contains Temporal data.
- **Text:** 46 columns contains textual information.
- **URLs:** 31 columns - Links to online resources etc...
- **Bool:** 18 columns - True/False values.
- **List:** 134 columns - List of numbers, categories or URLs etc...

Characteristics of Data

- The datasets vary in size, from small datasets like Pokemon Feature Correlation (1072 rows) to larger ones like US Migrations (288,300 rows).
- Several datasets are noisy and contain null values or redundancies, aiming to replicate real-world data.
- A simplified version of each dataset, DataBench_lite, is provided, containing only the first 20 rows for models that cannot handle large datasets.

Some of the Sample Datasets

- **Forbes Billionaires Dataset:** This contains business data with 2668 rows and 17 columns, containing information about the names and wealth of individuals.
- **Titanic Dataset:** This contains travel-related data with 887 rows and 8 columns and the targets predictions are for the survival.
- **NYC Taxi Dataset:** Contains 1M rows and 20 columns with trip details, distance, fares, and pickup/drop-off times.

Question and Answers

This dataset also contains 1300 manually generated Q-A pairs which are distributed across the datasets and are classified into 5 types: boolean, categorical, numbers, list of categories and a list of numbers.

The above analysis shows us that the dataset has included a wide range of domains and a wide range of data types. This property makes the dataset ideal for evaluating the models and the ability of the models to reason over diverse data formats.

Baselines

For the baseline model, we implemented a question-answering system designed to work with tabular data using a code generation approach. Our setup uses the codellama:7b model from the ollama library, which generates code snippets based on the question and dataset context. The system is designed to create Python functions that take pandas DataFrames as input and return the appropriate answer based on the question.

The baseline code can be accessed in the "ollama_codellama.py" file.

Model Architecture

The baseline architecture consists of the following components:

1. **Prompt Generator**: We generate prompts based on each row in the dataset, which includes the dataset information and the question. The function `example_generator` formulates these prompts and provides them as input to the codellama:7b model.
2. **Model Call**: The `call_ollama_model` function interacts with the locally hosted Codellama API. The prompts are sent via HTTP requests, and the model responds with Python code snippets that attempt to answer the question posed in the prompt.
3. **Postprocessing**: The generated code snippets are further processed using the `example_postprocess` function. This function attempts to extract the return statement from the model's response and evaluates it with the provided dataset. Any errors encountered during this step are logged as `__CODE_ERROR__`.

Prompt

```
# TODO: complete the following function in one line.
It should give the answer to: How many rows
are there in this dataframe?
def example(df: pd.DataFrame) -> int:
    df.columns=["A"]
    return df.shape[0]

# TODO: complete the following function in one line.
It should give the answer to: {question}
def answer(df: pd.DataFrame) -> {row["type"]}:
    df.columns = {list(df.columns)}
    return
```

Evaluation

We evaluated the baseline model using a subset of 320 examples from the `semeval` dataset's dev split. The dataset is loaded using the `utils.load_qa` method, and it is converted into a format compatible with our evaluation pipeline using the SemEval task default evaluation.

Performance

Using the evaluation framework, we got the following results:
Databench Accuracy: 15.6% Databench Lite Accuracy: 5%

Evaluation Scripts and Metrics

We will evaluate models using the metrics specified in the DataBench paper [1]. These include:

- **Accuracy**: Evaluate model performance based on different answer types and the number of columns in the dataset.
- **Exact Match (EM)**: Calculate the percentage of questions where the predicted answer exactly matches the ground truth.
- **F1 Score**: Provide a harmonic mean between precision and recall.
- **Mean Reciprocal Rank (MRR)**: Evaluate the rank position of the first correct answer.

Next Steps

- Refining prompting method to improve performance

- Finetune our models to improve code generation, specifically in our context of tabular data
- Explore other open-source model architectures other than codellama:7b, such as Llama 3.2, phi 3.5

References

[1] J. Osés Grijalba, L. A. Ureña-López, E. Martínez Cámara, and J. Camacho-Collados. Question answering over tabular data with DataBench: A large-scale empirical evaluation of LLMs. In N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13471–13488, Torino, Italia, May 2024. ELRA and ICCL.